

PROJECT BRIEF

LUCID

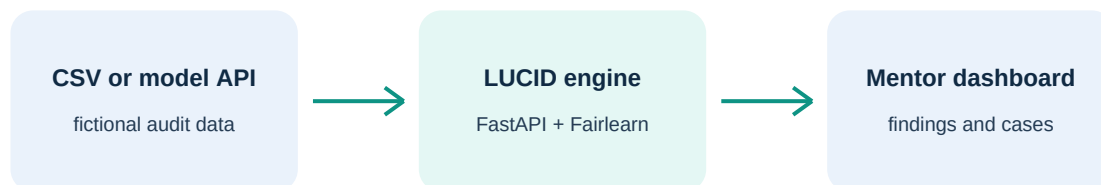
Mentor Demo Summary

A local-first, model-agnostic platform for assessing fairness in existing loan and hiring AI decisions.

Demo promise

LUCID assesses an existing AI system. It does not train, replace, or automatically modify that system.

LOCAL DATA FLOW



WHAT MENTORS WILL SEE

- A working local platform with fictional loan and hiring workspaces.
- Fairness findings supported by group metrics, confidence ranges, and policy evidence.
- A human-led review workflow with recommendations, case notes, notifications, and a print-ready report.

Core architecture

The first version avoids unnecessary cloud services, scheduling, and queues while retaining a production-shaped service boundary.

- **Next.js + TypeScript** provides the local dashboard for model setup, CSV mapping, findings, cases, and reports.
- **FastAPI + Fairlearn** provides validated decision and outcome APIs plus the model-agnostic fairness calculations.
- **PostgreSQL** stores models, events, immutable audit batches, policies, assessment runs, cases, and notifications.
- **Docker Compose** starts the web app, API, and database together on one laptop.
- **Demo Owner** is a real local user with both technical and compliance permissions for mentor demonstrations.

Data ingestion and audit trail

- Historical CSV uploads are optional. Organisations without history can begin with live API events and build evidence over time.
- Every model receives its own revocable API key. Each incoming event includes a pseudonymous source event ID and model version when available.
- Decision and outcome events are linked through the pseudonymous ID. Duplicate event IDs are safely ignored.
- Corrections append a new record rather than overwriting prior evidence. Late events create a revised assessment run with both timestamps preserved.
- Detailed event records are retained for 12 months. Audit summaries, policies, cases, and reports remain available afterward.

Fairness assessment policy

- LUCID initially supports binary business decisions: loan approve/reject and hiring shortlist/reject.
- Users select categorical fairness factors directly and define numeric bands such as age or income ranges. Free-text fields are excluded.
- LUCID compares selected groups without a privileged reference group and reports the largest observed gap.
- Intersectional analysis is opt-in and appears only when every combined group meets the sample-size requirement.
- Missing factor values remain visible as a data-quality cohort but do not trigger an alert unless explicitly included.

Editable policy default	Initial value	Purpose
Minimum decisions per group	100	Avoids very small comparisons
Decision-rate gap	15 percentage points	Flags approval or shortlisting disparity
Outcome-error gap	10 percentage points	Flags unequal error patterns
Outcome coverage	70% per group	Prevents premature error-rate claims
Confidence range	95%	Supports Needs review or Inconclusive wording

Privacy and evidence boundaries

- Only approved pseudonymous audit fields are stored. Direct identifiers, raw uploads, and unmapped source columns are not persisted.
- LUCID uses manual Run assessment actions only in the mentor demo; there is no daily automatic computation.
- LUCID labels findings Needs review or Inconclusive. It does not issue a legal conclusion that a model is unfair.
- Explanations describe observed evidence and possible contributors. They do not make unsupported claims about hidden black-box model reasoning.

Reviewer experience

- Two deterministic fictional workspaces - Northstar Lending and TalentBridge Recruiting - are ready to explore on first startup.
- The Demo Owner can add new AI models through the same industry-template, mapping, policy, and approval workflow used by the samples.
- The dashboard surfaces findings needing review first, then presents group metrics, confidence ranges, trends, coverage, and policy evidence.
- Automatic cases are created for policy breaches. Manual cases let reviewers document concerns found outside an automated alert.
- Every case supports evidence, possible contributors, recommendations, notes, a human decision, and closure.

Demo and delivery features

- Interactive local API documentation explains how an external model sends fictional decision, outcome, and correction events.
- A sample Python connector demonstrates an integration without exposing a real bank or recruiting system.
- In-app notifications and a local email preview demonstrate the notification flow without external email credentials.
- A print-ready internal report can be saved as PDF and remains linked to the immutable audit batch and policy version.
- Reset demo data restores the original fictional workspaces after practice runs.

First-version boundary

The mentor demo focuses on transparent assessment and human review. Scheduled monitoring, real email delivery, direct database connectors, ranking/text assessment, automatic mitigation, and production hosting are deliberate future extensions.

Technical implementation details

The first release is intentionally small, but its boundaries match the later production architecture.

- The web dashboard calls the FastAPI service over a local HTTP API. The API owns validation, access control, assessment runs, and audit writes.
- The Python assessment module is intentionally separate from the web interface so the same logic can later run behind scheduled workers or a hosted API.
- PostgreSQL is the system of record. It stores normalised entities and an audit-attribute JSON object so each model can use different approved factor names without schema rewrites.
- Docker Compose gives each service a named local network, persistent PostgreSQL volume, environment configuration, and one reproducible startup command.

Service	Responsibility	Local interface
Next.js dashboard	Model setup, CSV mapping, policy editing, findings, cases, reports	http://localhost:3000
FastAPI service	Validation, API-key checks, ingestion, assessment, OpenAPI docs	http://localhost:8000
PostgreSQL	Immutable entities, events, policy versions, findings, case history	Docker-only database port
Sample connector	Sends fictional decisions, outcomes, and corrections	Calls FastAPI locally

Core persisted records

- **Model:** workspace, industry template, display name, API-key hash, and current version metadata.
- **Audit batch:** immutable CSV import metadata, policy version, retention date, and assessment-run links.
- **Decision event:** model ID, source event ID, occurred time, received time, decision, optional score, version, and approved audit attributes.
- **Outcome and correction events:** append-only records linked to the original source event ID.
- **Assessment, finding, and case:** metric evidence, confidence range, status, recommendations, notes, and closure trail.

API contract and assessment lifecycle

- Each request authenticates with the registered model API key. The API resolves the model before accepting a decision, outcome, or correction event.
- The source event ID is unique within a model. Retried requests return the original acceptance result rather than adding a duplicate decision.
- CSV mapping transforms selected columns into the same canonical records used by the API. Raw source files are not retained after the immutable batch is created.
- Manual assessment runs freeze the applied policy version, selected factors, data cutoff, and calculation results for later comparison.

Endpoint	Purpose	Key fields
POST /v1/events/decisions	Ingest a binary business decision	source_event_id, occurred_at, decision, model_version, audit_attributes
POST /v1/events/outcomes	Attach a later business outcome	source_event_id, outcome, occurred_at
POST /v1/events/corrections	Append a correction without overwriting history	source_event_id, corrected fields, reason
POST /v1/assessments	Run the selected policy manually	model_id, factor selection, policy version, cutoff

Calculation logic and guardrails

- Selection rate is favourable decisions divided by decisions in each group. LUCID reports the largest group-to-group difference rather than comparing to a privileged reference group.
- When linked outcomes are available, error metrics are calculated only after each compared group reaches the configured 70% outcome coverage threshold.
- Bootstrap resampling creates 95% confidence ranges. A metric with too little reliable evidence is labelled Inconclusive rather than automatically opening an assertive finding.
- A Needs review case is created only when the configured sample size, coverage, confidence, and editable gap threshold are satisfied.
- The engine records metric values, group counts, confidence bounds, policy thresholds, and the data cutoff used in every result.

Black-box boundary

LUCID sees decisions, optional scores and outcomes, and approved audit factors. It can identify evidence patterns, but cannot infer hidden weights or prove causality without separate explainability evidence.